

Algebra 1 Unit 2 Assessment Guide

General Information

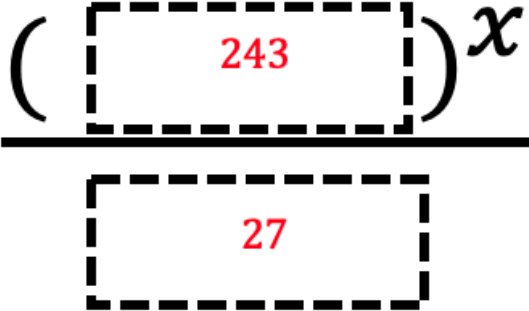
Although most of the questions on the unit assessment can be done without a calculator, it is suggested that students have access to a scientific calculator for the assessment. Teachers may choose to divide the assessment into two parts – a calculator and a no calculator assessment. That is left to each teacher's discretion.

The benchmarks on this assessment are closely related to benchmarks in grades 7 and 8. This means that if a question could assess the benchmark in grade 7 or grade 8, then it cannot be used in Algebra 1. The table provides further information on these relationships.

Grade 8 benchmark	Algebra 1 benchmark	Notes
MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.	MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.	- One of the grade 8 assessment limits states that the algebraic expression can have no more than two variables. This means that if a question has 3 variables, it would be considered Algebra 1. - Another grade 8 assessment limits states that if the algebraic expression has only one variable, then no more than three laws can be applied. This means that it is not likely that one variable could be used in Algebra 1 because the number of laws would have to be at least 4. - The last grade 8 assessment limit of note is that "items including two different variables are limited to the application of no more than two Laws of Exponents." This means that questions in Algebra 1 with two different variables will need to require the application of at least 3 different Laws of Exponents.
MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.	MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.	This means that questions that add or subtract linear expressions would be assessing MA.7.AR.1.1. This is why MA.912.AR.1.3 has the assessment limit "Items requiring the addition or subtraction of polynomial expressions must include at least one expression with a degree of two or more."

	MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.	The clarification of the grade 8 benchmark states that "Problems are limited to products where at least one of the factors is a monomial." Therefore Algebra 1 questions should not be a monomial times a binomial that has a degree of 1. This is why Algebra 1 has the assessment limit "Items requiring the multiplication of a monomial and a polynomial must include at least one expression with a degree of two or more."
	MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.	MA.912.AR.1.4 Divide a polynomial expression by a monomial expression with rational number coefficients.	The assessment limit for MA.912.AR.1.4 states "Polynomial expression given as the dividend is limited to two or three terms." This is because if the dividend is an expression of one term it would be assessing MA.8.AR.1.1, which is applying properties of exponents since essentially one term divided by one term only relies on laws of exponents.
	These subtle differences are important for the Algebra 1 teacher to know so that students are showing knowledge of benchmarks at the intended level. If teachers feel that their students are more capable of the content at the grade 7 or grade 8 level, then it is suggested that these students receive remediation and support. Teachers should consider how to monitor simple calculation errors as opposed to errors based on misconceptions of the benchmarks that are being assessed. If students have access to a scientific calculator, then assume that students are verifying their calculations with a calculator. Students commonly write x in the place of any variable. The expectation of an Algebra 1 student should be that they monitor themselves for these errors. The point deduction for students who repeatedly make this error should be minimal.		
Calculator Information	Although most of the questions on the unit assessment can be done without a calculator it is suggested that students have a scientific calculator. This allows the student to focus on showing their knowledge of the assessed benchmarks.		
Total Recommended Points	The total recommended points for the assessment are 96 points. Teachers may choose to distribute the extra 4 points or just round the total to 100.		

Question	1	Benchmark(s)	MA.912.NSO.1.2
	$180h^4m^{17}$	Recommended Scoring (6 Total Points)	Consider student points based on showing knowledge of application of each of the Laws of Exponents.
Question	2	Benchmark(s)	MA.912.NSO.1.2
	$-189x^{10}y^{-2}z^2$ or $-\frac{189x^{10}z^2}{y^2}$	Recommended Scoring (6 Total Points)	Consider student points based on applying the law of exponent to each variable.
Question	3	Benchmark(s)	MA.912.NSO.1.2
	$640h^6k^{-4}$ or $\frac{640h^6}{k^4}$	Recommended Scoring (6 Total Points)	Consider student points based on showing knowledge of application of each of the Laws of Exponents.
Question	4	Benchmark(s)	MA.912.NSO.1.2
	$2c^7$	Recommended Scoring (6 Total Points)	Consider student points based on showing knowledge of application of each of the Laws of Exponents.

Question	5	Benchmark(s)	MA.912.NSO.1.2
		Recommended Scoring (6 Total Points)	No partial credit is suggested.
Question	6	Benchmark(s)	MA.912.NSO.1.2
		Recommended Scoring (6 Total Points)	Consider 3 points for each correct value.
Question	7	Benchmark(s)	MA.912.AR.1.3
$2x^6 + 13x^5$		Recommended Scoring (6 Total Points)	Consider 3 points for correctly applying the distributive property and 3 points for adding like terms.

Question	8	Benchmark(s)	MA.912.AR.1.3
$k^2 - 3k - 88$		Recommended Scoring (6 Total Points)	Consider 3 points for correctly applying the distributive property and 3 points for adding like terms. Students who give the answer as $k^2 - 88$ should not receive any points.
Question	9	Benchmark(s)	MA.912.AR.1.3
$-5x^2 - 35x + 56$		Recommended Scoring (6 Total Points)	Consider 2 points for each correct application of the distributive property and 2 points for adding like terms.
Question	10	Benchmark(s)	MA.912.AR.1.3
$y^3 + \frac{2}{3}y^2 - 6y - \frac{17}{3}$		Recommended Scoring (6 Total Points)	Consider 2 points for correctly applying the distributive property to $7y^3 - 9y$, 2 points for correct distribution of the negative in front of $\frac{11}{3}y^3 - \frac{2}{3}y^2 + \frac{17}{3}$, and 2 points for adding like terms.
Question	11	Benchmark(s)	MA.912.AR.1.3
$10x^2 + 29x - 72$		Recommended Scoring (6 Total Points)	Consider 3 points for correctly applying the distributive property and 3 points for adding like terms. Students who give the answer as $10x^2 - 72$ should not receive any points.

Question	12	Benchmark(s)	MA.912.AR.1.3
	$x^2 + 14x + 49$	Recommended Scoring (6 Total Points)	Consider 3 points for correctly applying the distributive property and 3 points for adding like terms. Students who give the answer as $x^2 + 49$ should not receive any points.
Question	13	Benchmark(s)	MA.912.AR.1.3
	$\frac{44}{5}b^7 - 12b^6$	Recommended Scoring (6 Total Points)	Consider 3 points for correctly applying the distributive property and 3 points for adding like terms.
Question	14	Benchmark(s)	MA.912.AR.1.4
	$-3h^6 - 4h^{12}$	Recommended Scoring (6 Total Points)	Consider 3 points for correctly dividing each term.
Question	15	Benchmark(s)	MA.912.AR.1.3
	$x^2 - 11x + 28$	Recommended Scoring (6 Total Points)	Consider 3 points for correctly applying the distributive property and 3 points for adding like terms. Students who give the answer as $x^2 + 28$ should not receive any points.
Question	16	Benchmark(s)	MA.912.AR.1.4
	$4w^4y^8 + 7wy^2 - 9w^2$	Recommended Scoring (6 Total Points)	Consider 2 points for correctly dividing each term.